The signature operator on Witt spaces: Gysin maps and applications

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Pseudodifferential Techniques in Singular Analysis, 3/9/2026.
Based on papers in collaboration with Pierre Albin, Markus Banagl and Eric Leichtnam.

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- If $Y_\alpha, Y_\beta \in \mathfrak{S}$ and $Y_\alpha \cap \overline{Y_\beta} \neq \emptyset$, then $Y_\alpha \subset \overline{Y_\beta}$.
- Each stratum Y is endowed with a set of control data
$$T_Y, \quad \pi_Y : T_Y \to Y, \quad \rho_Y$$
Here T_Y is a neighbourhood of Y in $\widehat{X}$, $\pi_Y : T_Y \longrightarrow Y$ is a continuous retraction and $\rho_Y : T_Y \to [0, 2)$ is a radial function such that $\rho_Y^{-1}(0) = Y$.

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- The control data satisfy a set of compatibility axioms.
- From the axioms one can prove that for each $Y \in \mathfrak{S}$, $\pi_Y : T_Y \to Y$ is a locally trivial fibration with fibre the cone $C(L_Y)$ over some other ("simpler") stratified space L_Y
$$C(L_Y) \to T_Y \xrightarrow{\pi_Y} Y$$
L_Y is called the link over Y.

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- we set $X := \widehat{X} \setminus X_{n-2}$ and called it the regular part.

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- $\widetilde{X}$ has an additional structure: it has an an iterated fibration structure on the boundary (Melrose + Albin-Leichtnam-Mazzeo-P)
- This means that the boundary hypersurfaces are fibrations + compatibility relations at the corners between these fibrations.
- The interior of $\widetilde{X}$ is diffeomorphic to the regular part X of $\widehat{X}$.

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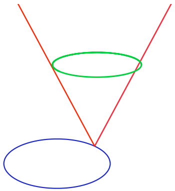

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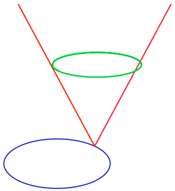

So there is a decomposition of $\widehat{X}$ into two **strata**: $\widehat{X} = Y \cup X$
NB: recall that strata are always **smooth** !.

- Y is the singular stratum (the bottom blue circle)
- X is the regular part (the **union** of the red cones (without the vertices)).
- The **link** of a point $p \in Y$ is a smooth closed manifold **Z** (the **green "circle"**).

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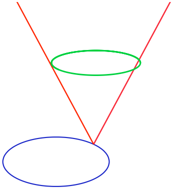

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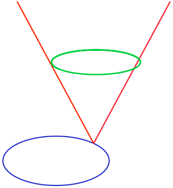

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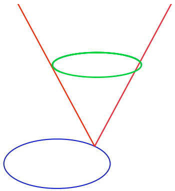

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Summarizing in the depth-one case the resolution is a manifold $\widetilde{X}$ with **fibered** boundary (over Y) and the boundary fibration $\partial(\widetilde{X}) \longrightarrow Y$ is precisely the link fibration $Z - H \longrightarrow Y$.

Wedge metrics and wedge Dirac operators.

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Consider $g_{H/Y}$, a **vertical** metric on the link-fibration $Z - H \longrightarrow Y$.

A wedge metric on X is an incomplete metric of the form

$$g := dx^2 + x^2 g_{H/Y} + \pi^* g_Y$$

near the singular stratum Y. Here g_Y is a metric on Y. This metric models a bundle of cones over Y.

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Near the singularity we can write D in the following form

$$cl\,(dx)\,\partial_x + \frac{1}{x}D_{H/Y} + \widetilde{\partial}_Y$$

where $\widetilde{\partial}_Y$ is a horizontal operator and where $D_{H/Y}$ is a vertical **family** of order 1 differential operators on the link-fibration $H \to Y$. D is an example of wedge differential operator.

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and notice that $xD = c\ell\,(dx)\,x\partial_x + D_{H/Y} + x\widetilde{\partial}_Y$ defines on $\widetilde{X}$ a edge differential operator, i.e. defined in terms of the Lie algebra of vector fields $\mathcal{V}_{\mathrm{e}}(\widetilde{X})$ given by

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Indeed in local coordinates $(x, y_1, \ldots, y_h, z_1, \ldots, z_v)$ with z_j coordinates on the fibers, the Lie algebra $\mathcal{V}_{\mathrm{e}}(\widetilde{X})$ is locally spanned by

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We define the order k wedge differential operators as

$$\mathrm{Diff}^k_{\mathrm{w}}(\widetilde{X}) := x^{-k}\mathrm{Diff}^k_{\mathrm{e}}(\widetilde{X})$$

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Crucial in the proof is the analysis of the wedge normal operator (really, the wedge normal family)

$$\mathcal{N}_y(\eth) = c\ell\,(dx)\,\partial s + \frac{1}{s}\eth_{Z_y} + \eth_{\mathbb{R}^v} \quad y \in Y$$

on $\mathbb{R}^+ \times Z_y \times T_yY \equiv \mathbb{R}^+ \times Z_y \times \mathbb{R}^v$, $v = \dim Y$, with a Vertical APS domain.

If the analytic Witt condition is satisfied this operator is invertible $\forall y$ and the inverse family can be used on the edge double space in order to build a parametrix with compact remainders.

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- Important: notice that there is no Hirzebruch (cohomological) L-class on a Witt space (we do not have a tangent bundle)

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Example 2. Certain quotients by non-free actions of Lie groups are Witt spaces.
For example, a semifree S^1-action on a smooth manifold M produces a singular quotient with links $\mathbb{C}P^N$.
If N is odd we get a Witt space.

The signature operator

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- Best treatment is due to Albin-Gell Redman; they developed a general edge calculus on manifolds with fibered corners.

A crash course in KK-theory

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- we shall be interested in analytic K-homology $K^a_*(-)$
- $K^a_*(X) := KK_*(C(X), \mathbb{C})$
- can define more generally $KK_*(A, \mathbb{C})$, A a C^*-algebra
- can define even more generally $KK_*(A, B)$ with B also a C^*-algebra
- we shall be interested in $KK_*(X, \mathbb{C})$ and $KK_*(C(X), C(Y))$ with X and Y Thom-Mather (compact) stratified spaces
- let us see definitions and key examples

$KK_*(A, \mathbb{C})$ definitions

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- a graded Fredholm module is a triple (H, ϕ, F) with $H = H^+ \oplus H^-$ a $\mathbb{Z}_2$ graded Hilbert **space**, $\phi : A \to \mathcal{B}(H)$ a representation, $\phi = \phi^+ \oplus \phi^-$, and $F : H \to H$ an **odd** bounded operator

$$F = \begin{pmatrix} 0 & F^+ \\ F^- & 0 \end{pmatrix}.$$

with the property that

$$(F^2 - 1)\phi(a)\,, \quad [F, \phi(a)]\,, \quad (F - F^*)\phi(a)$$

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  with the property that

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  are all **compact** operators
- we consider unitary classes of such triples
- we impose a equivalence relation through operator-homotopy
- this is $KK_0(A, \mathbb{C})$; for $KK_1(A, \mathbb{C})$ forget the grading !
- there is also an unbounded picture of $KK_*(A, \mathbb{C})$

Key example

Key example

- $X^{2\ell}$ is a smooth orientable compact m. without boundary
- g is a riemannian metric
- we consider $H := L^2(X, \Lambda^* X, g)$, the Hilbert space of L^2-forms
- $L^2(X, \Lambda^* X, g) = L^2(X, \Lambda^+ X, g) \oplus L^2(X, \Lambda^- X, g)$ (self dual and anti-self dual forms)
- $D_X^{\mathrm{sign}} \equiv D_X$ the signature operator (depends on g)
- it is $\mathbb{Z}_2$-graded **odd** with respect to $\Lambda^{\pm} X$
- let $\phi : C(X) \to \mathcal{B}(L^2(X, \Lambda^* X))$ be given by **multiplication**
- $\left(L^2(X, \Lambda^* X), \phi, D_X/(1 + D_X^2)^{1/2}\right)$ defines an element $[D_X]$ in $KK_0(C(X), \mathbb{C})$ in the **bounded picture**
- $\left(L^2(X, \Lambda^* X), \phi, D_X\right)$ defines the same element $[D_X]$ in $KK_0(C(X), \mathbb{C})$ in the **unbounded picture**.
- proof based on **classic elliptic theory**
- if X is odd dimensional, then the **odd** signature operator D_X defines $[D_X] \in KK_1(C(X), \mathbb{C})$
- they do not depend on the metric

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- need notion of **pre-Hilbert B-module**: a right B-module E equipped with a B-valued inner product $\langle\ ,\ \rangle : E \to B$ with natural properties ($\langle x, yb\rangle = \langle x, y\rangle b$ etc etc)

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- $KK_0(A, B)$ is defined as equivalence classes of (A, B)-Kasparov modules, i.e. triples (E, ϕ, F) where now:
  - $E = E^+ \oplus E^-$ is a $\mathbb{Z}_2$ graded Hilbert B-**module**,
  - $\phi : A \to \mathbb{B}(E)$ is an even homomorphism
  - $F \in \mathbb{B}(E)$ is odd and satisfies that

  $$(F^2 - 1)\phi(a)\,, \;\; [F, \phi(a)]\,, \;\; (F - F^*)\phi(a)$$

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The unbounded picture. Kasparov product

- we have explained the original definition with bounded operators
- there is again a definition through **unbounded** operators
- this is the so called **unbounded picture** of KK-theory
- one of the main result in KK-theory is the existence of the Kasparov product

$$KK_i(A,B) \times KK_j(B,C) \longrightarrow KK_{i+j}(A,C)$$

Second key example

Second key example

- $p : M \to B$ is a fiber bundle of smooth orientable compact manifolds without boundary
- let F be the fiber, of even dimension
- $g_{M/B}$ is a vertical riemannian metric
- $E^\infty := C^{\infty,0}(M, \Lambda^* M)$ is in a natural way a pre-Hilbert $C(B)$-module with

  $$\langle\, , \rangle : E^\infty \times E^\infty \to C(B)$$

  given by vertical integration
- let E be the completion of E^∞
- $C(M)$ acts on E by multiplication, call it ϕ
- let $D = \{D_b\}_{b \in B}$ be the vertical family of signature operators along the fibers
- then (E, ϕ, D) in an unbounded $(C(M), C(B))$ Kasparov module and defines a class

  $$\Sigma(p) \in KK_0(C(M), C(B))$$

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- $\Rightarrow$ there exists a well defined K-homology class $[D_g^{\mathrm{sign}}] \in K_*^a(\widehat{X}) \equiv KK_*(C(\widehat{X}), \mathbb{C})$ which is independent of g
  We adopt the notation $[D^{\mathrm{sign}}] \in K_*^a(\widehat{X})$

The K-homology class defined by $[D^{\mathrm{sign}}]$ on Witt spaces

- recall that we have a good elliptic theory for D^{sign}_g on a Witt space $\widehat{X}$ based on the edge pseudodifferential calculus
- $\Rightarrow$ there exists a well defined K-homology class $[D^{\mathrm{sign}}_g] \in K^a_*(\widehat{X}) \equiv KK_*(C(\widehat{X}), \mathbb{C})$ which is independent of g
  We adopt the notation $[D^{\mathrm{sign}}] \in K^a_*(\widehat{X})$
- Very important for us: recall the Chern character $\mathrm{Ch}_* : K_*(-) \to H_*(-)$.
  Moscovici-Wu: $\mathrm{Ch}_*[D^{\mathrm{sign}}] = L^{\mathrm{GM}}_*(\widehat{X})$ in $H_*(\widehat{X})$

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  with $\widehat{X}$ a Witt space. It is a sort of **orientation class** in the singular context.(I will not write $\otimes\mathbb{Z}[\frac{1}{2}]$ (but it is there)).
- $\Delta_{\widehat{X}}$ is the topological analogue of our $[D^{\mathrm{sign}}_{\widehat{X}}] \in K^a_*(\widehat{X})$
- he then studies systematically transfer (or Gysin or wrong-way) homomorphisms in topological K-homology

Topological Gysin maps

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- Banagl considers a fiber bundle $\widehat{X} \xrightarrow{p} \widehat{B}$ of Witt spaces, with **smooth** fiber F of dimension f and shows that if the structure group is a compact Lie group, then there exists a Gysin map

$$p^! : KO_*^{\text{top}}(\widehat{B}) \to KO_{*+f}^{\text{top}}(\widehat{X})$$

  such that

$$p^!(\Delta_{\widehat{B}}) = \Delta_{\widehat{X}}$$

- also, if $i : \widehat{Y} \hookrightarrow \widehat{X}$ is a codimension ℓ normally non-singular inclusion of Witt spaces, then there exists

$$i^! : KO_*^{\text{top}}(\widehat{X}) \to KO_{*-\ell}^{\text{top}}(\widehat{Y})$$

  such that $i^!(\Delta_{\widehat{X}}) = \Delta_{\widehat{Y}}$
- normally non-singular means that there exists a normal bundle

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- Question 1: can we prove these results for our class $[D^{\rm sign}_{\widehat{X}}]$?
- Question 2: can we generalize these results, allowing for example arbitrary fiber bundles of Witt spaces ?
- Question 3: can we compare the two classes, $\Delta_{\widehat{X}}$ and $[D^{\rm sign}_{\widehat{X}}]$ precisely ?

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- Question 3: can we compare the two classes, $\Delta_{\widehat{X}}$ and $[D^{\mathrm{sign}}_{\widehat{X}}]$ precisely ?
- Question 4: can we compare the topological and the analytic Gysin maps ?
- Short answers: yes !
  Our main tool is KK-theory and the Kasparov product.

Results

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Theorem

(Albin-Banagl-P. 2025, available in arXiv)

- *Given a fiber bundle of Witt spaces* $\widehat{F} - \widehat{X} \xrightarrow{p} \widehat{B}$ *there exists a class* $\Sigma(p) \in KK_f(C(\widehat{X}), C(\widehat{B}))$ *defined by the family of signature operators along the fibers*
- *we can define an analytic Gysin map* $p^! : K^a_b(\widehat{B}) \to K^a_{b+f}(\widehat{X})$ *by taking the left Kasparov product with* $\Sigma(p) \in KK_f(C(\widehat{X}), C(\widehat{B}))$
- $p^![D^{\mathrm{sign}}_{\widehat{B}}] = [D^{\mathrm{sign}}_{\widehat{X}}]$ *(Gysin preserves the signature class)*
- *if* $i : \widehat{Y} \hookrightarrow \widehat{X}$ *is a codimension* ℓ *normally non-singular inclusion of Witt spaces then there exists a class* $\Sigma(i) \in KK_\ell(C(\widehat{Y}), C(\widehat{X}))$
- *left Kasparov multiplication by* $\Sigma(i)$ *defines an analytic transfer map* $i^! : K^a_*(\widehat{X}) \to K^a_{*+\ell}(\widehat{Y})$
- $i^![D^{\mathrm{sign}}_{\widehat{X}}] = [D^{\mathrm{sign}}_{\widehat{Y}}]$ *(Gysin preserves the signature class)*

Results, part 2

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Theorem

(Albin-Banagl-P. 2025, available in arXiv) If $\widehat{X}$ is an n-dimensional Witt space and $c : KO_n^{\mathrm{top}}(\ \ ) \to K_n^{\mathrm{top}}(\ \ )$ is induced by complexification, then there exists an explicit automorphism $\alpha : KO_n^{\mathrm{top}}(\ \ ) \to KO_n^{\mathrm{top}}(\ \ )$ such that

$$c \circ \alpha(\Delta_{\widehat{X}}) \in K_n^{\mathrm{top}}(\widehat{X}) \otimes \mathbb{Z}[\frac{1}{2}]$$

corresponds to

$$2^{-\frac{n}{2}}[D_{\widehat{X}}^{\mathrm{sign}}] \in K_n^{\mathrm{a}}(\widehat{X}) \otimes \mathbb{Z}[\frac{1}{2}]$$

under the natural isomorphism $K_n^{\mathrm{top}}(\ \ ) \otimes \mathbb{Z}[\frac{1}{2}] \to K_n^{a}(\ \ ) \otimes \mathbb{Z}[\frac{1}{2}]$ (more on this isomorphism in a momemt).

Results, part 3

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- Consider again the generalized cohomology theory

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- Jakob has given a bordism-like description of the homology theory associated to a cohomology theory, in particular of $K^\bullet(B)[\frac{1}{2}]$
- we denote this homology theory as $K_\bullet^{b/2}(B)$
- Jakob proves that there exists an isomorphism $\varphi : K_\bullet^{b/2}(B) \to K_\bullet^{\mathrm{top}}(B)[\frac{1}{2}]$

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- we denote this homology theory as $K_\bullet^{b/2}(B)$
- Jakob proves that there exists an isomorphism $\varphi : K_\bullet^{b/2}(B) \to K_\bullet^{\mathrm{top}}(B)[\frac{1}{2}]$
- we prove, using the signature operator, that there exists an isomorphism $\kappa : K_\bullet^{b/2}(B) \to K_\bullet^{\mathrm{an}}(B)[\frac{1}{2}]$.
  This is the analogue of Baum-Higson-Schick for $K_\bullet^{\mathrm{geo}}(\ )$; our proof is different !

Results, part 3 (cont.)

Results, part 3 (cont.)

Consider the isomorphism

$$\lambda : KO_n^{\mathrm{top}}(B)[\tfrac{1}{2}] \to K_n^{\mathrm{an}}(B)[\tfrac{1}{2}]$$

obtained by composing the explicit automorphism

$$KO_n^{\mathrm{top}}(B)[\tfrac{1}{2}] \xrightarrow{\alpha} K_n^{\mathrm{top}}(B)[\tfrac{1}{2}]$$

with

$$K_n^{\mathrm{topo}}(B)[\tfrac{1}{2}] \xrightarrow{\varphi^{-1}} K_n^{b/2}(B) \xrightarrow{\kappa} K_n^{\mathrm{an}}(B)[\tfrac{1}{2}]$$

Theorem

(Albin-Banagl-P. May 2026; available in arxiv) Let $g : B \hookrightarrow Y$ be a normally nonsingular inclusion with oriented normal bundle ν of rank r. Then

$$\lambda \circ g^!_{\mathrm{top}} = g^!_{\mathrm{an}} \circ \lambda$$

Similar results hold for fibrations (with smooth fibers, since on the topological side only for these fibrations we have a Gysin homomorphism).

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- our 2 papers (Albin-Banagl-P) are 80+40 pp long; why ?! After all, we have all the work of Connes, Connes-Skandalis, Hilsum, Hilsum-Skandalis and many others.....
- Reason 1: Thom-Mather spaces are not stable under products. If $\widehat{X}$ and $\widehat{X}'$ are Thom-Mather spaces with stratifications $\mathfrak{S}$ and $\mathfrak{S}'$ and control data $\{T_Y, \pi_Y, \rho_Y\}_{Y\in\mathfrak{S}}$, $\{T'_Z, \pi'_Z, \rho'_Z\}_{Z\in\mathfrak{S}'}$ then $(\widehat{X}\times\widehat{X}', \mathfrak{S}\times\mathfrak{S}')$ with the product control data is NOT a Thom-Mather space.
- so, we had to give a new notion of smoothy stratified space, using conic charts, and develop from scratch the basics, for example the resolution of a stratified space to a manifolds with fibered corners.

Comments (cont.)

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- Reason 3: the edge pseudodifferential calculus takes place on the resolved manifold; there are complications in doing all this in the context of fiber bundles. In particular, starting from a fibration of Witt manifolds

$$\widehat{F} - \widehat{X} \longrightarrow \widehat{B}$$

we want a fibration of manifold with fibered corners where the base and the fibers are the resolutions of our Witt fiber $\widehat{F}$ and of our Witt base $\widehat{B}$. This is **not** the resolution of $\widehat{X}$.

A geometric application: the G-signature formula on Witt pseudomanifolds

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- we have seen that if $\widehat{F} \to \widehat{X} \xrightarrow{p} \widehat{B}$ is a fiber bundle of Witt spaces, then $\Sigma(p) \otimes [D^{\mathrm{sign}}_{\widehat{B}}] = [D^{\mathrm{sign}}_{\widehat{X}}]$
- this holds if $\widehat{X}$ is a real vector bundle over the Witt space $\widehat{B}$
- we use it crucially for a G-signature formula on Witt spaces
- if G is a compact Lie group then $\mathrm{Sign}(G, \widehat{X}) \in R(G)$ is defined using intersection homology;
- if $g \in G$ then taking characters we obtain $\mathrm{Sign}(g, \widehat{X}) \in \mathbb{R}$
- Atiyah-Segal-Singer gave a formula for $\mathrm{Sign}(g, X)$, when X is smooth
- proof based on the equivariant index theorem of Atiyah-Singer and the localization theorem in K-theory, due to Segal

G-signature formula on Witt spaces (cont)

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- let us recall Atiyah-Segal-Singer
- let $\mathcal{C}$ be the set of connected components of X^g and for $F \in \mathcal{C}$ denote by $N(F)$ the normal bundle of F in X
- then
$$\mathrm{Sign}(g, X) = \sum_{F \in \mathcal{C}} \langle \mathrm{AS}(N(F)) \cup L(F), [F] \rangle$$
where $\mathrm{AS}(N(F))$ is an explicit cohomological characteristic class associated to $N(F)$; $L(F)$ is Hirzebruch L-class of F

The G-signature formula on Witt spaces (cont)

The G-signature formula on Witt spaces (cont)

Theorem

(Banagl-Leichtnam-P., 2025; available in arxiv)
Let $\widehat{X}$ be a compact oriented Witt G-pseudomanifold. Assume that the inclusion $\widehat{X}^g \subset \widehat{X}$ is a normally nonsingular inclusion. Then

$$\mathrm{Sign}\,(g, X) \;=\; \sum_{F \in \mathcal{C}} \langle \mathrm{AS}(N(F))\,;\; L_*^{\mathrm{GM}}(F)\,\rangle$$

with $L_*^{\mathrm{GM}}(F) =$ Goresky-MacPherson- homology L-class of F.

The G-signature formula on Witt spaces (cont)

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- this is a neat formula in contrast with many contributions in G-index theory on singular spaces
- one proves that $F \subset X^g$ is indeed a Witt space
- remark: cannot use the equivariant A-S index theorem in the singular context
- fortunately Jonathan Rosenberg gave a KK-proof of the Atiyah-Segal-Singer G-signature formula and this is what we use here, together with our formula $\Sigma(p) \otimes [D_F^{\mathrm{sign}}] = [D_{N(F)}^{\mathrm{sign}}]$

The G-signature formula on Witt spaces (cont)

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The G-signature formula on Witt spaces (cont)

Ok, but do we have examples of normally non-singular inclusions of $\widehat{X}^g$ into $\widehat{X}$?! We have a very general transversality theorem. Here is a special case. Proofs are very delicate and use the whole Thom-Mather machinery.

Theorem

(Banagl-Leichtnam-P., 2025) Let M be a **smooth** *compact G-projective variety. Consider $Y \subset M$ a $G-$invariant* **singular** *subvariety. Assume that M^G is transverse to each stratum of Y. Then the inclusion of $Y^G = Y \cap M^G$ into Y is normally nonsingular.*

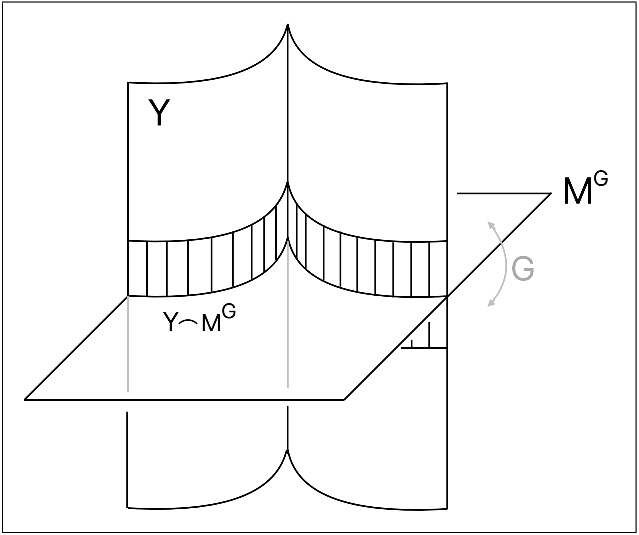

Figure: The transverse intersection of the stratified set $Y \subset M$ with the fixed point set of the ambient smooth action has an equivariant normal vector bundle neighborhood in Y, indicated by the shaded region.

THANK YOU